

CC- DISCRET12 | Discrete Structures 1

TOPIC	Handouts / Presentation Slides / Solution Sheets	Lecture Video
Module 0 - Course Introduction		
0.1 Class Onboarding	https://docs.google.com/presentation/d/1CJDS10CUjy7D9	
0.2 Course Guide	https://sherylsestella.notion.site/CC-DISCRET12-Course-Guide	
0.3 Virtual Class Rules & Online Etiquette	https://www.canva.com/design/DAEUuXWByw/AKjgCkd	
Module 1 - Set Theory & Its Applications		
1.1 Basic Concepts: Introduction to Set Theory		https://youtu.be/tyDKR4FG3Yw
1.1.1 Set Builder Notation & Roster Method	https://drive.google.com/file/d/1B1ogg3WZ7xRC5OMIQgC	https://youtu.be/FLgiccWl434
1.1.2 Subsets and Power Sets		https://youtu.be/H5D6EAezsXQ
1.1.3 Sets & Subsets Exercises		https://youtu.be/v818mV6jgBQ
1.2 Set Operations		https://youtu.be/4TlCToZZ5gA
1.2.1 Intersection of Sets, Union of Sets, and Venn Diagrams	https://drive.google.com/file/d/1bjraP1yOXgvvOuMiqk_Me	https://youtu.be/xZELQc11ACY
1.2.2 Symmetric Difference		https://youtu.be/-KQpyuaPs-E
1.2.3 Cartesian Product & Ordered Pairs		https://youtu.be/NnEkVooAsxk
1.2.4 Set Operations Examples		https://youtu.be/6RsudHXe6ZM
1.3 Applications of Sets	https://drive.google.com/file/d/1NiW9WmQbtFLvClfvyn7v	https://youtu.be/G8WDbo_o3So
Module 2 - Relations		
2.0 Relations - Presentation Deck	https://drive.google.com/file/d/1-8RjsNuDRPg9s3d82AnT	
2.0 Relations - Annotated Lecture Notes	https://drive.google.com/file/d/1TwlcemYe77NN48Gpb1W	
2.1 Introduction to Relations		https://youtu.be/qX3a7N1LyJg
2.2 Exercise 1 - Introduction to Relations		https://youtu.be/sDyBwYm2dAg
2.3 Properties of Relation		https://youtu.be/FjY0ciagOEE
2.4 Exercise 2 - Properties of Relation		https://youtu.be/fK_yqN5axr8
2.5 Operations on Relations	https://drive.google.com/file/d/1Pkftf9lqWjldyDzhmcD5Tf	https://youtu.be/3ANXg1JZ_2l
Module 3 - Logic Theory		
Webinar 1, Part 1 - Logic Theory	https://drive.google.com/file/d/1v0zWDbxhW2WtcKPlOtl 28027 CC-DISCRET12 Part 1 -->> 28027 CC-DISCRET12 Part 2 -->> 27805 CC-DISCRET12 Part 1 -->> 27805 CC-DISCRET12 Part 2 -->>	https://youtu.be/nmkcoF4vKsg https://youtu.be/1lm1fezW5XE https://youtu.be/uf3L5STHvbM https://youtu.be/8RpGKnm-NF8
Webinar 1, Part 2 - Logic Theory	https://drive.google.com/file/d/1olRETT5ZnYUMWbLYmNt 28027 CC-DISCRET12 -->> 27805 CC-DISCRET12 -->>	https://youtu.be/mr4LERIZPzg https://youtu.be/jDAjDCAUeck
Example 5 - Solutions Set	28027 CC-DISCRET12 -->> 27805 CC-DISCRET12 -->>	https://drive.google.com/document/d/1WwkOiztp7PCfjohw-zpwCB6QFFfM66a86ABx https://drive.google.com/document/d/1Vt1T834njwfp_U54NaMsyK0_LZPfjx7ydrbCf
Truth Tables - Examples	28027 CC-DISCRET12 -->> 27805 CC-DISCRET12 -->>	https://drive.google.com/sheets/d/1moph7kwmaLKW2j_ghi0Tt4-I9mkyRwbSXWd https://drive.google.com/sheets/d/1nnMYGh9p1rkbmt_VwhiZNFxmu7vZa6Z0Q91
Webinar 1, Part 3 - Logic Theory	28027 CC-DISCRET12 -->> 27805 CC-DISCRET12 -->>	https://youtu.be/WM2JbXVF-Bg https://youtu.be/DGKQa7AZU6U
Module 4 - Combinatorics		
Part 1 - Fundamental Principles of Counting	https://docs.google.com/presentation/d/1uknVY-Sh14w0c	https://youtu.be/IFJST9FnHcU
Fundamental Principles of Counting - Examples	https://drive.google.com/file/d/1KRuRXlib_sc83R1sr5Ynns	
Part 2 - Permutation & Combination	https://docs.google.com/presentation/d/1uknVY-Sh14w0c	https://youtu.be/jeqU6sOElv8
Examples: Permutation & Combination	https://docs.google.com/spreadsheets/d/1WnwKQ-worH1f	
Permutation & Combination Practice Problems	https://docs.google.com/spreadsheets/d/1Jjwb-2lxkcGUlX	
Combinatorics Decision Tree	https://miro.com/app/board/o9J_lFjR94M=?invite_link_id	https://youtu.be/uNiGeKOVuQA
Module 4 Wrap-up + Final Exam Briefing	28027 CC-DISCRET12 -->> 27805 CC-DISCRET12 -->>	https://youtu.be/mUve2kTdQ6c https://youtu.be/7j6d1819iv4

CC-QUAMETH21/22 | Quantitative Methods with Probability & Statistics

TOPIC	Handouts / Presentation Slides / Solution Sheets	Lecture Video
Module 0 - Course Introduction		
0.1 Class Onboarding	https://docs.google.com/presentation/d/1Z0yoVJjr4T7mwlBiwGL4BM	
0.2 Course Guide	https://sherylsestella.notion.site/CC-QUAMETH22-Course-Guide-d83	
0.3 Online Etiquette	https://www.canva.com/design/DAEUeuXWByw/AKjgCkdCSnfs3FN6	
Module 1 - Descriptive Statistics		
1.0 Overview of Statistics	https://miro.com/app/board/uXjVOJ9jqd8=?share_link_id=70153288	
1.1.1 Why you should love statistics? Alan Smith		https://youtu.be/ogeGJS0GEF4
1.1.2 Basic Concepts of Statistics	https://drive.google.com/file/d/1iFxdHgB7kw0IEQGwRk9p-m9rkG4l6	
1.2 Lecture: Collection, Organization and Presentation of Data	https://docs.google.com/presentation/d/11XvnfhKI--68kccuRxzymUU	https://youtu.be/LdPmkiQifYk
1.3 Lecture: Sampling Techniques	https://docs.google.com/presentation/d/1agHOPS-j3n4LHlvSEgHRT9	
Demo: Random Sampling using Table of Random Numbers	https://docs.google.com/spreadsheets/d/1_MNU07Sb59dQW9X6los	
Demo: Systematic Sampling	https://docs.google.com/spreadsheets/d/1qZPlLB9yW8Dfv-ijeiHqND4	
	Part 1 lecture video -->	https://youtu.be/3xTVbATmf_4?t=38
	Part 2 lecture video -->	https://youtu.be/9-FTRnx-s1U
1.4 Frequency Distribution	https://docs.google.com/presentation/d/1bBLjmmzTo2SDsRDR3nom	
Demo: Frequency Distribution Table	https://docs.google.com/spreadsheets/d/1HJR2ntas3sYFaG0UqVqOu	
	Part 1 lecture video -->	https://youtu.be/dtrs4E-2HD4
	Part 2 lecture video -->	https://youtu.be/kAT8Qz8OTsU
1.5 Lecture: Descriptive Measures	https://drive.google.com/file/d/1zrXqQJypNnbkg_XZSJ-fnlyzSQxY0PJ	
Miro board: Descriptive Measures	https://miro.com/app/board/o9J_IQDDiOo=?share_link_id=44816909	
Demo: Descriptive Measures	https://docs.google.com/spreadsheets/d/1LmH6itsyDRbZrEWiyVjvV	
Measures of Position	https://drive.google.com/file/d/19stmJPq6B0a364umqhlF7r_d4G_ryV	
	Part 1 lecture video -->	https://youtu.be/l2RzHdL_hAQ
	Part 2 lecture video -->	https://youtu.be/VpUQ0AQtnQE
Module 2 - Review on Combinatorics		
2.0 Combinatorics - presentation deck	https://drive.google.com/file/d/1BAP1J22QPH3NCEjbn91L7xWHXx	
2.1 Fundamental Principles of Counting Examples		https://youtu.be/0k88KKWTxVk
	Part 1 lecture video -->	https://youtu.be/e0dSEUP0Ang
	Part 2 lecture video -->	https://youtu.be/HUDLchcOlrc
2.2 Permutation & Combination Examples	https://drive.google.com/file/d/1Uy-15nfnMSHW-nQY-Zt_L8yeIlNcwc	
2.3 Combinatorics Decision Tree Diagram	https://miro.com/app/board/o9J_IFjR94M=?invite_link_id=41045909	https://youtu.be/uNiGeKOVuQA
Module 3 - Probability Theory		
Lecture: Probability Theory - Part 1	https://drive.google.com/file/d/1eGjn3m-Q-dWzL8aQAuMxHEXRmAw	https://youtu.be/0k88KKWTxVk
3.1 Classical Probability	https://drive.google.com/file/d/1PZ7-tpE0297D2jaECO_NCHRwoW9i	
3.2 Empirical Probability	https://drive.google.com/file/d/1j4F_ogniCY1A9cOoYxCyjGMhdFUfhC	
3.3 Mutually Exclusive Events	https://drive.google.com/file/d/14KwBjBUiEyRq3NS7PLUxGBUgLTrea	
3.4 Non-Mutually Exclusive Events	https://drive.google.com/file/d/11fJtcXyDohmJlMHnPiY5akXl1P_vJGk	
Lecture: Probability Theory - Part 2	https://drive.google.com/file/d/1eGjn3m-Q-dWzL8aQAuMxHEXRmAw	https://youtu.be/ViddOrMKJ6M
3.5 Conditional Probability	https://drive.google.com/file/d/1TmvcweaN_3v4IV2yIgiv8EU-RF2mj	
3.6 Independent Events	https://drive.google.com/file/d/1ai0E9gwMcZXXZo9OtWpQMRytFkh	
3.7 Dependent Events	https://drive.google.com/file/d/1TLhCbYE-bwvwlBBxAeWx3wETVAI	
3.8 Mathematical Expectations	https://drive.google.com/file/d/1gg4ctj-coD-wrhulV7cVzzN3gbEJ-wg	
Module 4 - Probability Distribution		
4.1 Probability Distribution - Part 1	https://drive.google.com/file/d/1uo1vwJ4RBZhtq60Tqin-c07yhqP1r2N	https://youtu.be/Vy00BIIFRJC

CC-QUAMETH21/22 | Quantitative Methods with Probability & Statistics

TOPIC	Handouts / Presentation Slides / Solution Sheets	Lecture Video
4.2 Probability Distribution - Part 2	https://drive.google.com/file/d/1uo1vwJ4RBZhtq6OTqin-c07yhgP1r2N	https://youtu.be/fUqBpdhXdbw
Terminal Group Project		
Project Briefing		https://youtu.be/i-CHVNuvvH0
Project Specifications	https://docs.google.com/presentation/d/1a6SBpFYmkFjbV7nnuoZxkl	
Social Media Survey	https://docs.google.com/document/d/1VHoLyP23gY7_0Nbpcam8LvR	
Blended Learning Experience Survey	https://docs.google.com/document/d/1GT5D2-53SiWnwgY9umO5Rp	
Rubrics for Grading	https://docs.google.com/document/d/19n4-DU2x3byxp0qYXFy3pF8	

CC-RESCOM31 Research Methods for Computing (BSIT Program)		
TOPIC	Handouts / Presentation Slides	Lecture Video
CLASS DIRECTORIES		
UC-CCS CAPSTONE PROJECTS INVENTORY	https://docs.google.com/spreadsheets/d/1968-dw8JNGb/	
Module 0 - Course Introduction		
0.1 Class Onboarding	https://docs.google.com/presentation/d/1mHyxjGF5hspN/ Recorded Video 04580 CC-RESCOM31 --> Recorded Video 15438 CC-RESCOM31 --> Recorded Video 30338 CC-RESCOM31 -->	https://youtu.be/bPnJltYiFM https://youtu.be/ujDggr9fe80 https://youtu.be/60myszZ6yrQ
0.2 Course Guide	https://sherylsestella.notion.site/CC-RESCOM31-Course-	
0.4 Virtual Class Rules & Online Etiquette	https://www.canva.com/design/DAEUeuXWByw/AKjgCkd/	
Module 1 - Context Setting		
1.1 Webinar 1 - Introduction to Capstone Project	https://drive.google.com/file/d/1_jliMpuRbqWtgJLr3snLV/ Recorded Video 04580 CC-RESCOM31 --> Recorded Video 15438 CC-RESCOM31 --> Recorded Video 30338 CC-RESCOM31 -->	https://youtu.be/cEv99s5QDAE https://youtu.be/cEv99s5QDAE https://youtu.be/CutiezGkXl0
Capstone Project Manual	https://drive.google.com/file/d/1_gZpkp1bQG0UwvXmyw6/	
Templates: Capstone Project Flow --> Capstone Project Manuscript Outline --> Capstone Project Manuscript Mechanics --> Software Requirements Specifications(SRS) -->	https://docs.google.com/document/d/1nYh7tasssZL_nPN/ https://docs.google.com/document/d/1tJtAvw9qK-MQ3Df/ https://docs.google.com/document/d/1_vxschR8YXp_Qpf/ https://docs.google.com/document/d/1-bQcggOybK445w/	
1.2 Webinar 2 - Startup in a Nutshell	https://drive.google.com/file/d/1-XSQNTLZ2s0fiUuJMAm/ Recorded Video 04580 CC-RESCOM31 --> Recorded Video 15438 CC-RESCOM31 --> Recorded Video 30338 CC-RESCOM31 -->	https://youtu.be/G04W1fEs1m0 https://youtu.be/xjGAayKmjSY https://youtu.be/15HvqMp99As
1.3 Introduction to Research	https://drive.google.com/file/d/1f4raAmeTnWIT1vEX6BRQ/	
1.4 Types of Research Design	https://drive.google.com/file/d/1bGkE1apMpfmFNoc5mhC/	
Module 2 - The Research Problem		
Webinar 3 Lesson 2.1- Creating Startup Ideas	https://drive.google.com/file/d/1CIVHO0DY6CfiqTsaeP2b/ Recorded Video 04580 CC-RESCOM31 --> Recorded Video 15438 CC-RESCOM31 --> Recorded Video 30338 CC-RESCOM31 -->	https://youtu.be/l_UZofP3Qq4 https://youtu.be/NOdQl2W9cls https://youtu.be/SPVeBCa3BzU
Lesson 2.2 - The Research Problem	https://drive.google.com/file/d/1r3ff7fpgzdlwLHW8SvUu/	
Lesson 2.3 - The Research Title	https://drive.google.com/file/d/1efqRaY_y8JC8p4NuMMr/	
Module 3 - Review of Related Literature & Studies		
3.1 Introduction to Review of Related Literature & Studies	https://docs.google.com/document/d/13NFnUVr5a8cSUd/	
3.2 How to Write Chapter 2	https://docs.google.com/presentation/d/1hVaswY7jfrWLz/	https://youtu.be/lKl1VsDh3bUY
Macro-level Summary of Related Studies Matrix	https://docs.google.com/document/d/1J5Wb2z1Pgk9NJKS/	
3.3 How to insert citation using MS Word		https://youtu.be/ZeFK3xVgAfU
Module 4 - Problem Validation		
4.1 Webinar 4 - Stage 1 Problem Validation	https://drive.google.com/file/d/1eMZGir2lj4eH3D4XpUjv6/ Recorded Video 04580 CC-RESCOM31 --> Recorded Video 15438 CC-RESCOM31 --> Recorded Video 30338 CC-RESCOM31 -->	https://youtu.be/x41oVgxAF7s https://youtu.be/832QbDFguc4 https://youtu.be/vE8STKWIHlw
4.2 Webinar 5 - Customer Insighting & Making Sense of your Research Findings	https://drive.google.com/file/d/1Ff5Ue2DOjsuajq0AlB35Z/ Recorded Video 04580 CC-RESCOM31 --> Recorded Video 15438 CC-RESCOM31 --> Recorded Video 30338 CC-RESCOM31 -->	https://youtu.be/SSR9MNOpgQ0 https://youtu.be/ly5t9HmMWdU https://youtu.be/yypovH3PCQs
Stage 1 - Problem Validation		
S1 Validation Manual	https://docs.google.com/document/d/1mjPgk2Qswld1cxP/ Recorded Video 04580 CC-RESCOM31 --> Recorded Video 15438 CC-RESCOM31 --> Recorded Video 30338 CC-RESCOM31 -->	https://youtu.be/srN6j4RYm_k https://youtu.be/nUfj8BdyQwY https://youtu.be/zeX2YlgsJX0
S1 Validation Report -outline	https://docs.google.com/document/d/1VeOD80GdrPQEYr/	

CC-RESCOM31 | Research Methods for Computing (BSIT Program)

TOPIC	Handouts / Presentation Slides	Lecture Video
Module 5 - Writing Chapter 1		
Webinar 6-Writing Chapter 1	https://drive.google.com/file/d/1JT4X4FQ-aAPQsF65MRh <i>Recorded Video 04580 CC-RESCOM31 --></i> <i>Recorded Video 15438 CC-RESCOM31 --></i> <i>Recorded Video 30338 CC-RESCOM31 --></i>	https://youtu.be/SzfDNFbiJM https://youtu.be/W0ho2byRyBM https://youtu.be/ntFeEbjWT9Y
Downloadable Materials: Sample: Rationale of the Study --> Sample: Significance of the Study --> Sample: Definition of Terms -->	https://drive.google.com/file/d/1jYTESsndLbz_4jFJdUCKV https://drive.google.com/file/d/1Af9-35ul4wckD3od84un https://drive.google.com/file/d/1nZgAwXHPNvX_cLwbqu	
Module 6 - The Other Parts of the Capstone Project Manuscript		
6.1 Gap Analysis	https://drive.google.com/file/d/1PynwpBw2Jn6Fg83Yxpm	
6.2 Webinar 7 - Concluding your Capstone Project Manuscript	https://drive.google.com/file/d/1KYs8az1QU2YAFFjPTZAC <i>Recorded Video 04580 CC-RESCOM31 --></i> <i>Recorded Video 15438 CC-RESCOM31 --></i> <i>Recorded Video 30338 CC-RESCOM31 --></i>	https://youtu.be/Rfb8DP5bmK8 https://youtu.be/YN9HJt_UShA https://youtu.be/RtXN5cXoDDA
Downloadable Materials: Capstone Project Manuscript Outline --> Title Hearing Manuscript Outline --> Capstone Project Manuscript Mechanics -->	https://docs.google.com/document/d/1tJtAvw9qK-MQ3Df https://docs.google.com/document/d/1ebmNhrNpWxfVb https://docs.google.com/document/d/1_vxschR8YXp_Qpf	
Final Briefing		
Final Briefing presentation deck	https://docs.google.com/presentation/d/1WhlToFZYrgVg	https://youtu.be/HtB5h2b9M-E
Downloadable Materials: Title Hearing Manuscript -outline --> Capstone Project Manuscript -mechanics --> Stage 1 Validation Manual --> Validation Report -outline --> Title Hearing Scoresheet -->	https://docs.google.com/document/d/1ebmNhrNpWxfVb https://docs.google.com/document/d/1_vxschR8YXp_Qpf https://docs.google.com/document/d/1mjPgk2Qswld1cxP https://docs.google.com/document/d/1VeOD80GdrPQEYg https://docs.google.com/spreadsheets/d/1qYJZHxUnWfF	

CC-TECHNO32 | Technopreneurship

TOPIC	Handouts / Presentation Slides / Worksheets / Reading Materials	Lecture Video
COURSE REQUIREMENTS		
1. CC-TECHNO32 Team Registration Form 2. Stage 2 - Solution Concept Validation Report Briefing: Stage 2 - Solution Concept Validation Stage 2 Validation Report - outline 3. Peer Evaluation 4. Final Startup Pitch Briefing: Final Startup Pitch Startup Entry - template Startup Entries - sample 5. Team Workbook	https://forms.gle/WYhsnyhd5IN7ikhA9 https://docs.google.com/document/d/1poQd6jl3SBhDV42ttWTyVwgu4t9 https://docs.google.com/document/d/1gpNFRjNoE5wMW-5mZLdRMideb https://forms.gle/xBmLqjWk5QxNKDCY6 https://docs.google.com/presentation/d/1ssUSQ4DLuF9_YTQ5Sf7g0fER https://docs.google.com/spreadsheets/d/1WeuB56tetbaHJd9mTw_PSVjN https://docs.google.com/document/d/1KbQEW3Zs2Ub0FZ09cn8xwOTrs https://docs.google.com/spreadsheets/d/1NKhER-B2WGL6UuqXI-F3z2Q	https://youtu.be/XNRULKhd3GM
MODULE 0 - COURSE INTRODUCTION		
0.1 Class Onboarding Guidelines for Team Formation & Startup Ideas Capstone Projects 2021 -2022 Curated Startup Resources 0.2 Course Guide 0.3 Virtual Class Rules and Online Etiquette	https://docs.google.com/presentation/d/1Ed9uWmT3Tlfl6ie4kfy_atiLRHf https://docs.google.com/presentation/d/1HT_udWIM0e-5jxsl7Zt46c-FI85 https://docs.google.com/spreadsheets/d/1X7Za2UV1jIjGuFjCA9wwrU7U https://bit.ly/sbsstartuplinks https://sherylsestella.notion.site/CC-TECHNO32-Course-Guide-41221945 https://www.canva.com/design/DAEUeuXWByw/AKjgCkdCSnfs3FN6ErW	
MODULE 1 - CONTEXT SETTING		
Lecture 1 - Startup in a Nutshell Design Thinking Design Thinking Case Study Lean Startup Case Study Startups and Innovation Success Stories: Amazon Google Netflix	Lesson 1-Startups in a Nutshell.pdf	https://youtu.be/ixQUCiKRkkg https://youtu.be/_r0VX-aU_T8 https://youtu.be/MkLc_TJldfw https://youtu.be/-_MH8TENpwc https://youtu.be/WGsMQLmyjM0 https://youtu.be/szbC4mZiVdQ https://youtu.be/M7qfeK6ntsE
PRELIMINARIES		
About the Preliminaries: Ideally, the students of Technopreneurship should have taken a pre-requisite course where they had to identify problem/s to solve and validate the problem. At the end of this pre-requisite course, they should have a validated problem and a startup idea based on the validated problem. In UC-Main, the said pre-requisite subject is CC-RESCOM31 for BSIT program. Bringing in the validated problem from CC-RESCOM31 to Technopreneurship class where the students will identify customer segments, validate the solution concept, and design the business model will provide enough time for them to go through the process of startup building. However, if the above case is not possible because there's no pre-requisite subject where students learned to identify and validate the problem(s) they want to solve and ideate startup solution based on the validated problem, then I suggest that the following Preliminaries should be included in Technopreneurship before proceeding to Module 2- Customer Discovery.		
Creating Startup Ideas Stage 1 Problem Validation Customer Insighting & Making Sense of your Research Findings S1 Validation Manual S1 Validation Report - outline	https://drive.google.com/file/d/1CIVH00DY6CfiqTsaeP2b2nJDxhDlccet/vi 4.1 Webinar 4- Stage 1 Problem Validation.pdf https://drive.google.com/file/d/1Ff5Ue2DOjsuajq0AIB35ZKOyOW3SAZM https://docs.google.com/document/d/1mjPgk2Qswld1cxPcbWhrPHLWZW https://docs.google.com/document/d/1VeOD80GdrPQEYpXNmImaXmtPo	https://youtu.be/L_UZofP3Qq4 https://youtu.be/x41oVgxAF7s https://youtu.be/SSR9MNOpqQ0 https://youtu.be/srN6j4RYm_k
MODULE 2 - CUSTOMER DISCOVERY		
2.1 Lecture 2 - Market Segmentation	https://drive.google.com/file/d/1v_Lit-eWykWN2PcB9rMMNQvqFoKShoc 17699 CC-TECHNO32 --> 17889 CC-TECHNO32 --> 26401 CC-TECHNO32 -->	https://youtu.be/7bQJq7LzpNo https://youtu.be/8DXOZTOVkkc https://youtu.be/CE-NdA_XciO
2.2 Basics of Customer Discovery	https://www.marketwithagility.com/blog/what-is-customer-discovery-and	

CC-TECHNO32 | Technopreneurship

TOPIC	Handouts / Presentation Slides / Worksheets / Reading Materials	Lecture Video
2.3 Phases of Customer Discovery	https://learn.marsdd.com/article/developing-your-value-proposition-an-o	
2.4 Desk Market Research	https://drive.google.com/file/d/1CKMADgpGYnz8YAkGoEbFCA88ssRa414	https://youtu.be/lhptmejZOBY
2.5 Lecture 3: Stage 2 Customer Validation	https://drive.google.com/file/d/1ZKvWncYB8lVTSAeoK1FLyira6U-6PHcW	https://youtu.be/6o9NtLBcUoo
Extra Learning Resources on Customer Validation:		
20 Questions You Can Ask to Validate Your Startup Idea	https://www.entrepreneur.com/article/241122	
What Grace Learned From Her Failed Startup		https://www.youtube.com/watch?v=Lbn76p_v
20 Reasons Startups Fail	https://drive.google.com/file/d/1qoWK8g_w2H4OqK9RbX7rs7Lgm3_C-B	
28 Tips for Creating Great Qualitative Surveys	https://www.nngroup.com/articles/qualitative-surveys/	
Better User Research Through Surveys	https://uxmastery.com/better-user-research-through-surveys/	
The UX of user research surveys	https://www.invisionapp.com/inside-design/ux-user-research-surveys/	
Nancy Duarte: How to Create Better Visual Presentations	https://youtu.be/so9EJoQJc-0	
Data Viz Project	https://datavizproject.com/	
95 Ways to find your first customers for customer development or your first sale	https://jasonvanish.com/2013/08/11/95-ways-to-find-your-first-customers-for	
9 Free-to-Use Wireframing Tools	https://www.interaction-design.org/literature/article/10-free-to-use-wireframing	
24 Ways to Test your Business Ideas	https://drive.google.com/file/d/1N4HIV1kiCP9HJD7Xm1QBx7-Kx9RHK-1/vie	
Sample Experimentation Plan	https://drive.google.com/file/d/1Ys7DS-Oa2dpudNz94AGnom05cyfFL9qq/vie	
MODULE 3 - PROTOTYPING		
3.0 Prototyping - presentation deck	https://drive.google.com/file/d/1zZ3HHJsv05jZvZ1L7B2tygYUPOAJyE3/v	
3.1 Introduction to Prototyping		https://youtu.be/l-tR0Xd9og4
3.2 Common Methods of Prototyping		https://youtu.be/QDZtyUss8wc
3.3 How test a prototype?		https://youtu.be/w9sMPu6NNaE
3.4 Examples of Prototype		
Fanbaro	https://drive.google.com/file/d/1jTGGJ6NU030wMQ3E4ccmPVNSaPi3dS	
Guia	https://drive.google.com/file/d/18w6djf3ss1YxYPIKXSiRciFHLqDKBUP/v	
VenturePal	https://drive.google.com/file/d/1EdBIDUnP3xRwYjEvJFZUn1dpp3CLys4/v	
MODULE 4 - BUSINESS MODELING		
4.1 Practical Finance	https://drive.google.com/file/d/1-tSXDxybA00Tzvf47fENcVECGvqqsut/v	https://youtu.be/_X9Dks3Hzlo
Demo: Practical Finance	https://docs.google.com/spreadsheets/d/1Sym4_ydXBxhVAWPMRay5w8	
Required Readings:		
Competitor-based Pricing	https://www.profitwell.com/recur/all/competitor-based-pricing	
Customer Lifetime Value	https://clevertap.com/blog/customer-lifetime-value/	
Making Freemium Work	https://hbr.org/2014/05/making-freemium-work	
Market Sizing	https://www.mindtools.com/pages/article/market-sizing.htm	
Value-based Pricing	https://hbr.org/2016/08/a-quick-guide-to-value-based-pricing	
4.2. Business Model Innovation: Amazon, Spotify, Tinder		https://youtu.be/avWVPaJFgFk
4.2.1. Lecture: Business Modeling	https://drive.google.com/file/d/1foOivJYmxw0VKV6vGFgkmVuAaYDXbfD	https://youtu.be/Wga3foue8eA
4.2.2. Business Modeling using the Lean Canvas:		
An Introduction to Lean Canvas	https://medium.com/@steve_mullen/an-introduction-to-lean-canvas-5c17	
Lean Canvas Example - Uber	https://youtu.be/pvIN9STpzCQ	
Lean Canvas Example - Airbnb	https://youtu.be/Csofgp9Bcyo	
Lean Canvas Guide	https://drive.google.com/file/d/1bmSwYOmC_Rz5vwV7yWmeE4Snv7Px9	
Lean Canvas Template	https://docs.google.com/spreadsheets/d/1veK3eZbSo0qVNWBJnTFLVEkd	
Lean Canvas Poster	https://drive.google.com/file/d/1UKJBCpWJL98yJ5tXzEDMY9R0WmBYGJ	

CC-TECHNO32 | Technopreneurship

TOPIC	Handouts / Presentation Slides / Worksheets / Reading Materials	Lecture Video
4.2.3. Business Modeling using the Business Model Canvas: Business Model Canvas Explained Building Blocks of Business Model Canvas Business Model Canvas Example - Uber Business Model Canvas Example - Airbnb Strategyzer Webinar: Ways To Present The Business Model Canvas Business Model Canvas Guide Business Model Canvas Template Business Model Canvas Poster	https://www.strategyzer.com/business-model-canvas/building-blocks https://drive.google.com/file/d/1v3QmaVGEs2Vhh-CS1Fd9REZlec79gnc/ https://docs.google.com/spreadsheets/d/1e99kDNI11vWla25W78JvRe9h0/ https://drive.google.com/file/d/14j-6rPMAntJHYTsA30PQ0lyHicSQsKy-/v	https://youtu.be/QoAOzMTLP5s https://youtu.be/dtf61YDCXDY https://youtu.be/WzGhFmiB9G0 https://youtu.be/SliMK6Z2jIA
4.2.4 Designing your Startup's Value Proposition: Value Proposition Canvas - Explained Value Proposition Canvas Example - Uber Value Proposition Canvas Trigger Questions Value Proposition Canvas Instruction Manual Value Proposition Canvas Template Value Proposition Canvas Poster	https://drive.google.com/file/d/1cUvgXi3FzCLjuhJbn0Pn_FoiKYNbdDCA/v https://drive.google.com/file/d/1PZgf1rs_B23EvfteoVd6OCUblVU9RrJM/v https://docs.google.com/spreadsheets/d/1NpJ03mPNQjJ7UvMh4BcDQkV/ https://drive.google.com/file/d/1EZmrDb4gsWuq9WQlElyN6XZQOEzPzb5/	https://youtu.be/ReM1uqmVfP0 https://youtu.be/kiWkRiynPAo
4.3. Revenue Models: 11 Types & How To Pick The Right One?	https://www.profitwell.com/recur/all/11-popular-types-of-revenue-models	
4.4 Nine Business Models and Metrics Investors Want		https://youtu.be/PTg3RZPXgLG
MODULE 5 - MINIMUM VIABLE PRODUCT		
5.1 Lecture 4: Introduction to Minimum Viable Product	https://drive.google.com/file/d/1E3ZsxRZHgsxA9TMOjJcM4BIWvBqmVH/	https://youtu.be/Z-QZISZJb5c
5.2 Building a Product		https://youtu.be/C27RVio2rOs
MODULE 6 - STARTUP PITCH		
6.1 Lecture 5: Startup Pitch Pitch Planner Canvas	https://drive.google.com/file/d/14Jagax8efoJTpXxiqsnGLKDEdc293Bih/v https://docs.google.com/spreadsheets/d/1P7N_skBCHPBKuSca4YOAx4r/	https://youtu.be/XUQ4YZ-lpds?t=61
6.2 Startup Pitch Examples: Tourmo FindN SQUIDEW Hoomle Solar Hope SmarterMeter		https://youtu.be/gfpDvDHP7sw https://youtu.be/atxx6sl4mPQ https://youtu.be/ujd6pQCm4mY https://youtu.be/VjnFwx5gkIg https://youtu.be/crAvbk72Tqo https://youtu.be/-cgWqf_mv2A
6.3 The early pitch decks of 29 startups before they became billion-dollar companies	https://www.cbinsights.com/research/billion-dollar-startup-pitch-decks/	

IT-FRELEAN Lean Startup Methodology		
TOPIC	Handouts / Presentation Slides / Worksheets / Reading Materials	Lecture Video
COURSE REQUIREMENTS		
1. Team Registration	https://forms.gle/fnzo89Ab5TMUvX948	
2. Final Startup Pitch	https://docs.google.com/presentation/d/1VVA7wrPMvQm5M7J8zAoej8_tf5	
3. Peer Evaluation	https://forms.gle/2dja23EqY44wXX5v9	
4. Stage 3 - MVP Validation Report	https://docs.google.com/document/d/1ONZSEmP2RY5UUAXBwpsfg2YwWp	
5. Team Workbook	https://docs.google.com/spreadsheets/d/1suYyTAB6fJgUVMMSiBWq1gDLq	
MODULE 0 - COURSE INTRODUCTION		
0.1 Class Onboarding	https://docs.google.com/presentation/d/1hN6GKi5v248uu00E5igLY9zEkST	
0.2 Course Guide	https://sherylsestella.notion.site/IT-FRELEAN-Course-Guide-cf41f4c5fad2	
0.3 Virtual Class Rules and Online Etiquette	https://www.canva.com/design/DAEUeuXWByw/AKjgCkdCSnfs3FN6ErWYc	
MODULE 1 - CONTEXT SETTING		
1.1 Lean Startup Methodology	https://drive.google.com/file/d/1X7DfwWzqrslwm2x3lKbS_Q5PXweojB9Xv	https://youtu.be/RSalOCHbuYw
1.1.1 Lean Startup Lecture by Eric Ries		https://youtu.be/PXUd7wXrb0Q
1.1.2 Lean Startup Lessons for Founders		https://youtu.be/CYeTNRIQYKk
1.2 Startup Playbook	https://playbook.samaltman.com/	
1.3 Lecture 1- Innovation Lifecycle and Stages	https://drive.google.com/file/d/1TKqCs0HmkdTy6LU9Rlczhjd_5BatgMdv/vi 27359 IT-FRELEAN 07823 IT-FRELEAN 36061 IT-FRELEAN	https://youtu.be/PNsMT_29WJE https://youtu.be/RujJEOC7iGY https://youtu.be/lBOU-fRq4Cs
MODULE 2 - MINIMUM VIABLE PRODUCT		
2.1 Review: Introduction to Minimum Viable Product	https://drive.google.com/file/d/1-qla-S-8FYHlARUC1Gg_2vKrAxU_Pccx/vie	https://youtu.be/Z-QZlSZJb5c
2.2 Lecture 3 - Planning & Testing your MVP	https://drive.google.com/file/d/14RHH-XSRaOm3YA9EX9_anDDcBjxmIYxb 27359 IT-FRELEAN --> 07823 IT-FRELEAN -->> 36061 IT-FRELEAN-->>	https://youtu.be/6oTfDGfmx0g https://youtu.be/4pbstJwB8bY https://youtu.be/gm_GLnIEbJU
2.3 Building Product		https://youtu.be/C27RVio2rOs
2.4 Additional Resources on MVP:		
How Uber, Airbnb & Dropbox Released MVPs to Achieve Rapid Growth	https://medium.com/@LoganTjm/how-uber-airbnb-dropbox-released-mvps-	
3 Awesome Minimum Viable Products		https://youtu.be/xPJog_QVsY4
How to get and test startup ideas?		https://youtu.be/vDXkpJw16os
Validation Guide: 24 Ways to Test your Business Idea	https://drive.google.com/file/d/1_5RBt7RyoM12uUeS3CeYev8iUULaJieo/vie	
MODULE 3 - KEY METRICS		
3.1 Lecture 4 - Startup Metrics	https://drive.google.com/file/d/1mb90lP_-4djgKgnD2nHN-q7u4BWbTVtZv 27359 IT-FRELEAN 07823 IT-FRELEAN 36061 IT-FRELEAN	https://youtu.be/KCP7tAX_k98 https://youtu.be/L3x-MuWDIzg https://youtu.be/14J8VrO4-Lo

IT-FRELEAN Lean Startup Methodology		
TOPIC	Handouts / Presentation Slides / Worksheets / Reading Materials	Lecture Video
3.2 Nine Business Models and the Metrics Investors Want		https://youtu.be/PTg3RZPXgLG
3.3 How to Set KPIs and Goals	https://www.ycombinator.com/library/6j-how-to-set-kpis-and-goals-sus-20	
3.4 16 KPIs to help your business succeed	https://www.geckoboard.com/blog/the-a-z-guide-to-startup-metrics-16-kpi	
3.5 The Value of Keeping the Right Customers	https://hbr.org/2014/10/the-value-of-keeping-the-right-customers	
MODULE 4 - MARKETING & BRANDING		
4.1 Marketing Strategy	https://drive.google.com/file/d/1Z3DJNzmTl9ri_YUHV_2tk3nThyw1VYsj/view	https://youtu.be/wG1h7bqNF60
4.2. What is a Marketing Plan & How to Write One?	https://blog.hubspot.com/marketing/marketing-plan-examples	
4.3 Customer Retention	https://blog.hubspot.com/topic-learning-path/customer-retention	
4.4 How to Build a Brand	https://drive.google.com/file/d/1Pf_OGePMOwe2tC7J7QK1wYexX2UTj-HH/view	
4.5 What is a brand book and how to create one?	https://www.designrush.com/trends/brand-book	https://youtu.be/ADGamW-ZKe0
4.6 Brand Book Examples	https://thenextweb.com/news/a-look-at-brand-books-from-the-best-in-the-world	
Additional Resources	https://docs.google.com/document/d/1lghH8h37n3A3n2m3AylsLv_2W7o-C/view	
MODULE 5 - STRATEGY & IMPACT		
5.1 Lecture 6 -Startup Roadmapping & Impact	https://drive.google.com/file/d/1S1NFtuO10Hijk-OPeTv6i7JFQ29b4zd/view	
Examples of Culture Deck:		
Netflix Culture	https://jobs.netflix.com/culture	
Zappos Culture	https://www.zappos.com/about/what-we-live-by	
Module 6 - Fundraising		
6.0 Fundraising presentation deck	https://drive.google.com/file/d/1DbVNXU-EEJGFxQLKutzXM7T6eiTSnrWo/view	https://youtu.be/QUUswPcflI4
6.1 Lecture 7 - Fundraising Part 1		
6.2 Lecture 8 - Fundraising Part 2		
6.2 Key Terms to Include in Founders' Agreement	https://startupguide.hbs.edu/people/founding-team/key-terms-to-include-in-agreement	
6.3 How to Split Equity Among Co-Founders	https://www.ycombinator.com/library/5x-how-to-split-equity-among-co-founders	https://youtu.be/adpQQosyNGo?t=510
6.4 Required Readings on Fundraising	https://docs.google.com/document/d/1UazdWxHQhFxacX306d0vmWfJ8O8/view	

CC-TWRITE21 Technical Writing & Presentation Skills		
TOPIC	Handouts / Presentation Slides / Reading Materials / Worksheets	Lecture Video
Module 0 - Course Introduction		
0.1 Class Onboarding	https://docs.google.com/presentation/d/1IVKEf_OILfyCLNSN3QMIZynIE 06924 CC-TWRITE21 30262 CC-TWRITE21 18762 CC-TWRITE21 35741 CC-TWRITE21 07104 CC-TWRITE21	https://youtu.be/Fj2IgLQ5nxY https://youtu.be/kMqSK42kf5Y https://youtu.be/a2QlbtESwgk https://youtu.be/BO6949IIT7g F2F class onboarding
0.2 Course Guide	https://sherylsestella.notion.site/CC-TWRITE21-Course-Guide-bfcc9f7b	
0.3 Virtual Class Rules and Online Etiquette	https://www.canva.com/design/DAEUeuXWByw/AKjgCkdCSnfs3FN6Er	
Module 1 - Fundamentals		
1.1 Technical Writing in a Nutshell	https://docs.google.com/presentation/d/1tEmnJzJbC9l6NJTcSFBwVUKr	https://youtu.be/ieC4rcvmy2k
1.2 5 Traits of Technical Writing	https://docs.google.com/presentation/d/1gNNtX4UXf0Dx6ByMMtuuqg	https://youtu.be/jSuoKZ_m1Hk
1.3 Elements of Technical Writing	https://docs.google.com/presentation/d/1x1LD3mnXuNlPOFqZQJRCw1E	https://youtu.be/BohVLB0bMFo
1.4 Lecture: The Writing Process	https://docs.google.com/presentation/d/1_lX4-cZdvoDz-o7WnlutVf6Yr 06924 CC-TWRITE21 30262 CC-TWRITE21 18762 CC-TWRITE21 35741 CC-TWRITE21 07104 CC-TWRITE21	https://youtu.be/k5qnkXF9nfM https://youtu.be/iduFx8GqPA8 https://youtu.be/p5gwxdlp4l0 https://youtu.be/gpJ-PDQwWcU https://youtu.be/hhA-JXlkQB0
1.5 Technical Research	https://docs.google.com/presentation/d/1untWab4Qw3ponlnnrreJqoXlp	
Technical Research -Part 1		https://youtu.be/_dW0xGXO7FU
Technical Research -Part 2		https://youtu.be/ZnEilijUZ8k
Technical Research -Part 3		https://youtu.be/gARJ7lveAQY
Technical Research -Part 4		https://youtu.be/e1UUqPVkSmU
1.6 Document Design & Graphics	https://docs.google.com/presentation/d/1FKeMWYJqaHYSvwF476XA2a	https://youtu.be/n0pKbPJX_a
1.7 Lecture: Document Design Elements	https://docs.google.com/presentation/d/14lqTs0gTgVwSyaeiz6Uzeij-86 06924 CC-TWRITE21 30262 CC-TWRITE21 18762 CC-TWRITE21 35741 CC-TWRITE21 07104 CC-TWRITE21	https://youtu.be/HgtoFSI_jo8 https://youtu.be/lQg18r5iGk0 https://youtu.be/Cp886wmeDf8 https://youtu.be/6kPFij5qprE N/A
Module 2 - Applications		
2.1 Writing Proposals	https://docs.google.com/presentation/d/1aUzhTYEoM3bEgS9J5alznZ-	https://youtu.be/D5YlfgGvDvg
2.2 Brief Correspondence	https://docs.google.com/presentation/d/1V3RONNYf_HeB1t5reci_wZ39	https://youtu.be/i64bhoQjt2c
2.3 Email	https://docs.google.com/presentation/d/1mCAklRAQR-q69dq6Mc9aj8v	

CC-TWRITE21 | Technical Writing & Presentation Skills

TOPIC	Handouts / Presentation Slides / Reading Materials / Worksheets	Lecture Video
Introduction to Email		https://youtu.be/SaEhJ45n50o
Planning your Email		https://youtu.be/mnOo0zR4Mvw
Writing your Email		https://youtu.be/9pMi76lyGD0
Email Etiquette		https://youtu.be/WAkxUbw6l4
Sending Emails: Best and Worst Practices	https://drive.google.com/file/d/1eFrtaYvYtAM2wWMzGaAifoM4_thQhJL	https://youtu.be/DO6CMKLxs2w
2.4 Letters	https://docs.google.com/presentation/d/1MDi3GjrbrFEYdPC8ug9PSA6e	https://youtu.be/Jb3YicZByw0
2.5 Memos	https://docs.google.com/presentation/d/1AFb7FF0zOQdzKzpRtxEynFyc	https://youtu.be/UvU9PGBYXdU
2.6 Lecture: Professional Branding	https://drive.google.com/file/d/1laxgOHo7ZafLNrah0Fm55xfNGypMoH	https://youtu.be/c7R2a4h_W9M
2.7 Creating a Professional Portfolio	https://docs.google.com/presentation/d/129tRPw21SBj50nAF0vjsClrM	https://youtu.be/pggdqh850cY
2.8 Writing a Professional Resume	https://docs.google.com/presentation/d/17FdUeC9VnLm88VltNLBVzin	https://youtu.be/NwkCjU9J158
2.9 Writing a Cover Letter	https://docs.google.com/presentation/d/1CdmSF5FTYPK-NX_AG3OPw	https://youtu.be/Sa_2mMVYyV4
2.10 Lecture: Fundamentals of Presentation	https://drive.google.com/file/d/1E_WarN02kcSVeKFQmgkphAtH1z5azV	
Major Projects		
Midterm Project	https://docs.google.com/document/d/1AnnHuYZaNqvmCjYhqdT24buXC	https://youtu.be/rb06ct--9JQ
Final Project	https://docs.google.com/document/d/1nzJNhpNvAXmse6G1zYLzEYs75V	

ENTREP101 | The Entrepreneurial Mind

TOPIC	Handouts / Presentation Slides / Worksheets / Reading Materials	Lecture Video
FINAL REQUIREMENTS		
1. Final Startup Pitch	https://docs.google.com/presentation/d/1NwyfLmoF-VGKTDGU6wbRUYpQx/	https://youtu.be/sngjepwWhz8
Startup Entry - template	https://docs.google.com/spreadsheets/d/1HIUZLCEYCOG4kXQ1rp7y2bQeupl	
Pitch Deck - template	https://docs.google.com/presentation/d/1gsHtBOWmxwmmeDIBWgpQpnjxsS0	
Sample startup entries	https://docs.google.com/document/d/16n0h16XpFny1oS1lFnsGnWBOJLsjCZ7	
2. BEEHIVE MOOC Certificate		
Module 0 - Course Introduction		
0.1 Class Onboarding	https://docs.google.com/presentation/d/1y970S6WLQcie4hEDf1iKoHkcsqq_Q	https://youtu.be/UPfwm758a5A
0.2 Course Guide	https://sherylsestella.notion.site/ENTREP101-Course-Guide-b2fde5d4d80346e	
0.3 Virtual Class Rules and Online Etiquette	https://www.canva.com/design/DAEUeuXWByw/AKjgCkdCSnfs3FN6ErWYoA/	
0.4 Step-by-Step Guide to BEEHIVE MOOC Registration	https://drive.google.com/file/d/14vqv8IRNHj-zjkbEMCSce2miOOvsPx6R/view	
0.5 BEEHIVE MOOC Registration Demo		
		https://youtu.be/MumUTLeOD5Q
Module 1 - Introduction to Entrepreneurship		
Units 1 - 4 (BEEHIVE MOOC)	http://beehive-erasmusplus.eu/courses/entrepreneurship-mooc/	https://youtu.be/DZTpyd0fPC0
Lecture 1 - Startups in a Nutshell	https://drive.google.com/file/d/1gONgK4GI0yYphGXWfGPylRAg1iYg3OBI/view	
Lecture 2 - How to create a startup idea?	https://drive.google.com/file/d/13r2pEMYejSN5DLk2O_AgiR5g33kegcSH/view	
Group Activity 1: Proposed Startup Idea	https://docs.google.com/document/d/1P6bLa0dHZyODFJhG1Q3OnchlsAKzN/	
Module 2 - Essentials in Startup Building		
Units 5 - 8 (BEEHIVE MOOC)	http://beehive-erasmusplus.eu/courses/entrepreneurship-mooc/	https://youtu.be/PTg3RZPXgLg
Lecture 3 - Market Segmentation: An Overview	https://drive.google.com/file/d/1u5fzQ6c6QLXqZi7CZ1FcUz4_j2CZCBAJ/view	
Group Activity 2: Customer Segmentation	https://docs.google.com/document/d/1XRoh3HV17symE6LsC-fv48WVxhej7ZC	
Lecture 4 - Desk Market Research	https://drive.google.com/file/d/1rWnJ-3iSm0oQhah_XmmxBJr-yawN5mL8/view	
Group Activity 3: Competitors' Analysis	https://docs.google.com/document/d/1FI9hLHe6r6vZqRQmYE-pznXPbjPZ10_	
Units 9 - 12 (BEEHIVE MOOC)	http://beehive-erasmusplus.eu/courses/entrepreneurship-mooc/	
Lecture 5: Customer Journey	https://drive.google.com/file/d/1ZqsJs1_NYUzqU_t0_8fl_5q1adZj5Zz52/view?u	
Group Activity 4: Customer Journey	https://docs.google.com/document/d/1Rafc623ALaKWIG8mAbWTchrFhM0LBv	
Lecture 6 - Lean Canvas	https://drive.google.com/file/d/1zyBGVvV3-paXy_3FZwXTAf6nVXMkDDPB/view	
Nine Business Models and Metrics that Investors Want		
Group Activity 5: Business Model	https://docs.google.com/document/d/1cb-v17Fqtvfr6K0pkJQYPZj7MZnngSLD/	

ENTREP101 | The Entrepreneurial Mind

TOPIC	Handouts / Presentation Slides / Worksheets / Reading Materials	Lecture Video
Module 3 - Entrepreneurial Management & Strategy		
Units 13 - 18 (BEEHIVE MOOC)	http://beehive-erasmusplus.eu/courses/entrepreneurship-mooc/	
Lecture 7 - Startup Pitch	https://docs.google.com/presentation/d/1deSsU9Y9-aELVEZC-t4m52nxPe4Jy	

CS-DIFFCAL21 | Differential Calculus (BSCS Program)

	Website	Solutions Set	Handouts	Video
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	Website	Solutions Set	Handouts	Video
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	Website	Solutions Set	Handouts	Video
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	Website	Solutions Set	Handouts	Video
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Module 0 - Course Introduction

0.1 Class Onboarding

<https://docs.google.com/presentation/d/1U8vYtXqZjK9mM>

<https://youtu.be/J2EnfIKP9mM>

0.1 Class Onboarding

<https://docs.google.com/presentation/d/1U8vYtXqZjK9mM>

<https://youtu.be/J2EnfIKP9mM>

0.1 Class Onboarding

<https://docs.google.com/presentation/d/1U8vYtXqZjK9mM>

<https://youtu.be/J2EnfIKP9mM>

0.2 Course Guide <https://sherylsestella.notion.site/C>

0.2 Course Guide <https://sherylsestella.notion.site/C>

0.3 Meet the Community

0.4 Online Etiquette <https://www.canva.com/design/DA>

0.4 Online Etiquette <https://www.canva.com/design/DA>

0.5 Expectations, Concerns & Expectations

Module 1 - Limits

1.1.1 Introduction to Limits <https://youtu.be/YNstP0ESndU>

1.1.1 Introduction to Limits <https://youtu.be/YNstP0ESndU>

1.1.2 The Functional Notation <https://drive.google.com/file/d/1RU>

1.1.2 The Functional Notation <https://drive.google.com/file/d/1RU>

1.2.1 Limit of a Function <https://drive.google.com/file/d/1c8>

1.2.1 Limit of a Function <https://drive.google.com/file/d/1c8>

1.2.2 Limit Laws <https://youtu.be/0By02ToZ7AU>

1.2.2 Limit Laws <https://youtu.be/0By02ToZ7AU>

1.2.3 Finding Limits Algebraically <https://youtu.be/CSALbevEMfs>

1.2.3 Finding Limits Algebraically <https://youtu.be/CSALbevEMfs>

1.2.4 Challenging Limits-Examples https://youtu.be/_wE4HxK-Ztc

1.2.4 Challenging Limits-Examples https://youtu.be/_wE4HxK-Ztc

1.3 Indeterminate Forms <https://drive.google.com/file/d/10N>

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Module 2 - Differentiation of Algebraic Functions

2.1 The Derivative of a Function <https://drive.google.com/file/d/1xB>

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Webinar 2-Lessons 2.3-2.6 <https://drive.google.com/file/d/19o> <https://youtu.be/T0RQnFQdgsA>

Webinar 2-Lessons 2.3-2.6 <https://drive.google.com/file/d/19o> <https://youtu.be/T0RQnFQdgsA>

Webinar 2-Lessons 2.3-2.6 <https://drive.google.com/file/d/19o> <https://youtu.be/T0RQnFQdgsA>

2.3 The Chain Rule <https://drive.google.com/file/d/16C>

2.3 The Chain Rule <https://drive.google.com/file/d/16C>

2.4 Differentiation of Inverse Function <https://drive.google.com/file/d/1yo>

2.4 Differentiation of Inverse Function <https://drive.google.com/file/d/1yo>

2.5 Higher Derivative <https://drive.google.com/file/d/1Y4>

2.5 Higher Derivative <https://drive.google.com/file/d/1Y4>

2.6 Implicit Differentiation <https://drive.google.com/file/d/1DF>

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CS-DISCRET21 | Discrete Structures 2 (BCS Program)

	Website	Handouts	Video
Module 0 - Course Introduction			
0.1 Class Onboarding		https://docs.google.com/presentation/d/1dQ8	https://youtu.be/Q7NMgoJO0ts
0.2 Course Guide	https://www.notion.so/sherylsestella/CS-DISCRET21		
0.3 Meet the Community			
0.4 Virtual Class Rules & Online Etiquette		https://www.canva.com/design/DAEUeuXWBy	
0.5 Expectations, Concerns & Expectations			
Module 1 - Boolean Algebra			
1.1.1 A Quick Overview of Boolean Algebra			https://youtu.be/cTjFY18SjRc
1.1.2 Boolean Functions		https://drive.google.com/file/d/1JruBDfyAQNG	
1.2.1 Understanding Logic Gates			https://youtu.be/iNEtYZqtjTo
1.2 Logic Gates		https://drive.google.com/file/d/1VcHpEzstRgl	
1.3.1 The Laws of Boolean Algebra			https://youtu.be/EPJf4owqwdA
1.3.2 Simplifying Boolean Expressions		https://drive.google.com/file/d/1CdpxHM1Pae	
1.3.3 Simplifying Complex Boolean Expressions			https://youtu.be/XMCW6NFLM8g
Module 2 - Graph Theory			
2.1 Webinar 1 - Introduction to Graph Theory		https://drive.google.com/file/d/1ve-4jGcEGPg	https://youtu.be/RsnaYmkYB2Q
2.2 Graph Terminology Part 2		https://drive.google.com/file/d/1-w_mXFhX05	
Optional Resources: Bipartite Graphs:			
Bipartite Graph			https://youtu.be/HqIUbSA9cEY
Complete Bipartite Graph			https://youtu.be/iKeA1c-bdXM
Webinar 2 - Graph Terminology Part 3 & Graph Representation			https://youtu.be/FQqrZww2uuY
2.3 Graph Terminology - Part 3		https://drive.google.com/file/d/1My57JQwvbZ	
2.4 Graph Representation		https://drive.google.com/file/d/1KmJJAKSibDA	
2.5 Shortest Path		https://drive.google.com/file/d/1i6C3BHrbzPr	
Module 3 - Trees			
Webinar 3 - Introduction to Trees		https://docs.google.com/presentation/d/16Xy	https://youtu.be/kqyDuc5YG20
Sample: Pre-order Traversal Method by Tonilon		https://docs.google.com/presentation/d/1Ruy	
Sample: In-order Traversal Method by Fabroa		https://drive.google.com/file/d/1LKnumX0sX3	
Sample: Post-order Traversal Method by Ampatin		https://docs.google.com/presentation/d/1rTIs	
Webinar 4 - Trees Part 2		https://docs.google.com/presentation/d/13g9	https://youtu.be/n00vYwui98e